# Human Trust Calibration in Human–Agent Collaboration for Autonomous Research

## Abstract

The rapid development of autonomous and agentic artificial intelligence (AI) is changing the nature of human–AI collaboration. Unlike conventional decision-support systems that primarily provide recommendations, autonomous research agents can plan tasks, retrieve information, analyze evidence, use external tools, generate outputs, and execute multiple steps with limited human intervention. These capabilities create significant opportunities for improving the speed and scale of research, but they also introduce an important human-factors problem: how much should humans trust and rely on autonomous AI agents? This assignment examines human trust calibration in human–agent collaboration, with particular attention to autonomous research. Drawing on research on trust in automation, human–AI interaction, algorithmic advice, explainable AI, and human–AI teamwork, the paper argues that effective collaboration requires calibrated trust rather than unconditional trust. Calibrated trust occurs when human reliance corresponds appropriately to an AI agent's actual capabilities and limitations. The paper discusses over-trust, under-trust, automation bias, transparency, explanations, reliability, uncertainty communication, prior experience, and human oversight. It concludes by proposing that autonomous research systems should be designed to continuously support trust calibration through transparent evidence provenance, uncertainty communication, performance feedback, appropriate human checkpoints, and adaptive levels of autonomy.

**Keywords:** trust calibration, human–AI collaboration, autonomous agents, agentic AI, autonomous research, appropriate reliance, automation bias, human–agent teaming

## 1. Introduction

Artificial intelligence is increasingly moving from passive information-processing tools toward autonomous agents capable of performing complex, multi-step tasks. In research environments, an AI agent may formulate search strategies, retrieve academic literature, compare sources, summarize findings, identify patterns, generate hypotheses, and support the production of research outputs. Such systems can potentially reduce researchers' workload and accelerate knowledge discovery. However, increased autonomy also changes the relationship between humans and AI. The human is no longer simply interacting with a software tool; instead, the human may supervise, delegate tasks to, evaluate, and collaborate with an artificial teammate.

A major challenge in this relationship is trust. **Lee and See (2004)** argue that trust influences how people rely on automation, particularly when the system is complex and humans cannot completely understand its operation. Importantly, effective human–

automation interaction does not require people to trust automation as much as possible. Rather, reliance should be appropriate to the actual capabilities of the system. This principle is particularly relevant to autonomous research agents because an agent may produce fluent, convincing, and apparently well-supported answers while still making errors in source selection, interpretation, reasoning, or citation.

Research on AI trust reinforces the importance of this issue. **Glikson and Woolley (2020),** in their review of empirical research, show that human trust in AI is influenced by characteristics of the AI system, the context of interaction, and how the AI is represented to users. **Hoff and Bashir (2015)** similarly distinguish between dispositional, situational, and learned forms of trust in automation. These perspectives indicate that trust is dynamic rather than a fixed characteristic of the human user.

Therefore, the central issue for autonomous research is not simply whether researchers trust AI, but whether their **trust is calibrated**. A researcher who accepts every AI-generated claim without verification demonstrates over-trust. Conversely, a researcher who rejects accurate AI assistance because of excessive skepticism demonstrates under-trust. Both conditions can reduce the potential benefits of human–agent collaboration.

**2. Understanding Trust Calibration**

Trust calibration refers to achieving an appropriate correspondence between a person's trust in an automated system and the system's actual capabilities or reliability. **Lee and See (2004)** describe appropriate reliance as a central objective of trust in automation. If a system is highly reliable, insufficient reliance can prevent users from benefiting from it. If a system is unreliable, excessive reliance can lead to errors.

This distinction is particularly important because trust and reliance are related but not identical concepts. Trust represents a person's attitude or expectation concerning the system, whereas reliance is reflected in actual behavior. A researcher may report high trust in an AI research agent but still independently verify every source. Alternatively, a researcher may express moderate trust while frequently accepting the agent's recommendations. Consequently, research on trust calibration should ideally measure both subjective trust and observable reliance.

**Hoff and Bashir (2015**) provide a useful framework by identifying three levels of trust: dispositional trust, situational trust, and learned trust. Dispositional trust concerns a person's general tendency to trust automation. Situational trust is affected by the immediate environment and task circumstances. Learned trust develops through experience with a particular system. In autonomous research, these factors may interact. For example, a researcher who is generally skeptical of AI may initially have low trust, but repeated successful interactions with a research agent may increase trust over time.

## 3. Trust in Autonomous Research Agents

Autonomous research agents differ from traditional automation because they may operate across several stages of a task. A conventional search engine responds to a query, whereas an autonomous research agent can potentially decide what information to search for, which sources to examine, how to synthesize evidence, and what additional searches are necessary.

This expanded autonomy creates several trust challenges.

First, the **complexity of agent behavior** can make it difficult for users to understand why a particular conclusion was produced. Second, agents may operate with incomplete or changing information. Third, generative AI systems can produce plausible but incorrect information. Fourth, autonomous agents can potentially propagate an initial error through several subsequent stages of a research workflow.

The problem becomes particularly serious when the agent's communication style creates an impression of competence that exceeds its actual reliability. Human users may interpret fluent language, confident explanations, or anthropomorphic interaction as evidence of intelligence and reliability. **Glikson and Woolley (2020)** show that the representation and communication style of AI can influence trust. Therefore, interface design is an important component of trust calibration.

## 4. Over-Trust, Under-Trust, and Appropriate Reliance

There are three broad outcomes of human trust calibration.

### 4.1 Over-trust

Over-trust occurs when users rely on an AI system beyond what its actual capabilities justify. In autonomous research, this could involve accepting AI-generated references without checking them, assuming that an AI-generated summary accurately represents the original paper, or allowing an agent to make methodological decisions without human review.

Over-trust can create automation bias. Users may accept an automated recommendation even when contradictory evidence is available. This is especially dangerous in research because errors can be incorporated into later analyses and eventually appear in published or submitted work.

### 4.2 Under-trust

Under-trust occurs when users fail to rely on an AI system despite evidence that it can perform a task effectively. This may lead researchers to ignore useful recommendations or unnecessarily repeat tasks manually.

**Dietvorst, Simmons, and Massey (2015)** demonstrated a related phenomenon known as **algorithm aversion**. People can become disproportionately reluctant to use algorithms after observing them make mistakes, even when the algorithm performs better overall than a human. This finding is important for autonomous research because a single visible AI failure may substantially reduce subsequent reliance.

### 4.3 Appropriate reliance

The desired condition is appropriate reliance. Users should accept AI assistance when the agent demonstrates competence and independently evaluate or reject its recommendations when the task exceeds its capabilities.

This idea is supported by research showing that people can also exhibit **algorithm appreciation**. **Logg, Minson, and Moore (2019)** found that people sometimes prefer advice when they believe it originates from an algorithm rather than a human. Together, algorithm aversion and algorithm appreciation demonstrate that human reliance can be biased in either direction.

## 5. Factors Affecting Trust Calibration

Several factors are particularly important in autonomous research.

### 5.1 Reliability and past performance

Repeated successful performance can increase trust, whereas failures can decrease it. However, users may not always update trust proportionally to actual performance. A highly visible failure can have a larger psychological effect than several successful interactions.

Therefore, autonomous research systems should provide users with meaningful information about performance rather than allowing trust to develop solely through subjective impressions.

### 5.2 Transparency and explanations

Transparency can help users understand how an AI system reaches its conclusions. **Amershi et al. (2019)** proposed 18 guidelines for human–AI interaction, emphasizing practices such as making system capabilities and limitations understandable and supporting appropriate user control.

However, explanations do not automatically produce calibrated trust. **Bansal et al. (2021)** found that explanations could increase people's acceptance of AI recommendations regardless of whether those recommendations were correct. This finding is especially significant for autonomous research: an explanation that sounds persuasive may increase trust without necessarily increasing accuracy.

Consequently, explanations should not merely make an AI system appear understandable. They should help users determine **when the system should and should not be trusted**.

### 5.3 Uncertainty communication

A research agent should communicate uncertainty when evidence is incomplete, conflicting, or weak. For example, instead of presenting a conclusion as definitive, an agent could distinguish between:

- well-established findings;
- findings supported by limited evidence;
- conflicting findings;
- interpretations made by the AI; and
- claims requiring human verification.

Such distinctions can prevent users from treating every AI output as equally reliable.

### 5.4 Source provenance

Source provenance is particularly important for autonomous research. Users should be able to identify where a claim originated, whether the source is peer-reviewed, how directly the source supports the claim, and whether the AI has inferred information beyond what the source states.

A research agent that provides traceable evidence can make human verification more efficient and can therefore support calibrated reliance.

### 5.5 Human expertise

The user's level of expertise can influence reliance. **Logg et al. (2019)** found that experienced professionals sometimes relied less on algorithmic advice than laypeople. In research, subject-matter expertise may therefore act as an important moderating factor. An experienced researcher may identify weaknesses in an AI-generated argument that a novice researcher overlooks.

This suggests that trust calibration should be personalized according to user expertise rather than assuming that all users interact with autonomous research agents in the same way.

## 6. Human Oversight in Autonomous Research

Increasing autonomy does not necessarily mean eliminating human involvement. **Shneiderman (2022)** argues for human-centered AI systems that amplify and augment human capabilities while maintaining meaningful human control.

For autonomous research, a useful model is **adaptive human oversight**. Routine and low-risk activities can be delegated to the agent, whereas high-impact decisions should require human review.

For example:

| **Research activity** | **Suggested AI autonomy** | **Human involvement** |
|---|---|---|
| Initial literature search | High | Review search strategy |
| Source categorization | High | Spot-check results |
| Literature summarization | Medium–High | Verify important claims |
| Evidence synthesis | Medium | Review reasoning |
| Research hypothesis generation | Medium | Human judgment required |
| Methodological decisions | Low–Medium | Human approval |
| Final conclusions | Low | Human responsibility |
| Academic citations | Medium | Verify every important citation |

This approach recognizes that different research tasks have different levels of risk. The appropriate level of autonomy should therefore depend on the consequences of an error rather than simply on what the AI is technically capable of doing.

**7. A Conceptual Framework for Trust Calibration**

A useful conceptual model for autonomous research can be expressed as:

**AI capability → Evidence of performance → Human perception of reliability → Trust → Reliance → Research outcome → Feedback → Trust recalibration**

The process should be dynamic rather than one-directional.

For example, suppose an autonomous research agent identifies ten relevant papers. The researcher examines the papers and discovers that eight are highly relevant and two are inappropriate. The researcher should update their trust based on this experience. If later interactions show improved performance, trust may increase. If the agent repeatedly provides unsupported claims, trust should decrease.

The system itself can support this process by providing performance indicators, confidence estimates, source provenance, explanations, and explicit warnings about limitations.

The overall principle can therefore be expressed as:

**Human reliance should be proportional to demonstrated AI competence and inversely related to uncertainty and task risk.**

This does not mean that users should calculate a numerical trust score for every interaction. Instead, the design of autonomous research systems should continuously provide information that enables humans to make better reliance decisions.

### 8. Challenges and Ethical Considerations

Trust calibration also raises ethical concerns. Excessive dependence on autonomous research agents may weaken researchers' critical-thinking skills and encourage the outsourcing of intellectual responsibility. If an AI agent makes an incorrect claim, responsibility cannot simply be transferred to the system.

Academic integrity is another concern. Researchers remain responsible for verifying evidence, accurately representing sources, and distinguishing original human reasoning from AI-generated material.

There is also a risk of **trust asymmetry**. An AI system may be highly trusted even when its limitations are poorly understood. Conversely, humans may be expected to supervise systems whose internal processes are too complex to meaningfully evaluate. Effective human–agent collaboration therefore requires not only better AI systems but also better human training.

Finally, trust calibration should not be treated as purely an interface problem. It is a sociotechnical issue involving system design, user expertise, organizational policies, accountability, task characteristics, and the consequences of errors.

### 9. Research Gap

Although trust in automation and trust in AI have been extensively studied, autonomous research agents create a relatively new research context. Much earlier research examined relatively constrained systems such as automated vehicles, decision-support systems, industrial automation, and robots. Autonomous research agents are different because they can perform flexible, knowledge-intensive, multi-step tasks.

A significant research opportunity therefore exists in studying how trust changes when an AI agent moves from providing a recommendation to independently planning and executing research activities.

Future studies could investigate questions such as:

1. How does increasing agent autonomy affect human trust?
2. Does providing source provenance improve trust calibration?
3. How do AI errors affect trust in multi-step autonomous research?
4. Do explanations improve appropriate reliance or simply increase acceptance?
5. How does researcher expertise influence reliance on autonomous research agents?
6. Can adaptive human checkpoints reduce over-reliance?
7. How should uncertainty be communicated during autonomous literature review?
8. Does long-term interaction produce appropriate trust or excessive dependence?

These questions could be investigated through controlled experiments comparing different levels of AI autonomy, explanation, uncertainty communication, and human oversight.

**10. Conclusion**

Human trust calibration is a central requirement for effective human–agent collaboration in autonomous research. The objective should not be to make researchers trust AI systems as much as possible. Instead, autonomous research agents should encourage **appropriate reliance**, in which human behavior corresponds to the agent's demonstrated competence, limitations, uncertainty, and the consequences of potential errors.

Previous research on automation demonstrates that both over-trust and under-trust can be problematic (**Lee & See, 2004**; **Hoff & Bashir, 2015**). Research on AI further shows that system characteristics, interaction context, explanations, and user characteristics influence trust (**Glikson & Woolley, 2020**; **Amershi et al., 2019**). Evidence concerning algorithm aversion and algorithm appreciation demonstrates that human reliance can be biased in opposite directions (**Dietvorst et al., 2015**; **Logg et al., 2019**). Furthermore, explanations may increase acceptance without necessarily improving the correctness of decisions (**Bansal et al., 2021**).

For autonomous research, calibrated trust should therefore be supported through transparent evidence provenance, uncertainty communication, performance feedback, appropriate explanations, adaptive autonomy, and meaningful human oversight. The most effective future research systems are unlikely to be those that replace researchers completely. Rather, they will be systems that **augment human researchers while helping them understand when to rely on AI, when to question it, and when to take control themselves**.